Deblending random seismic sources via independent component analysis

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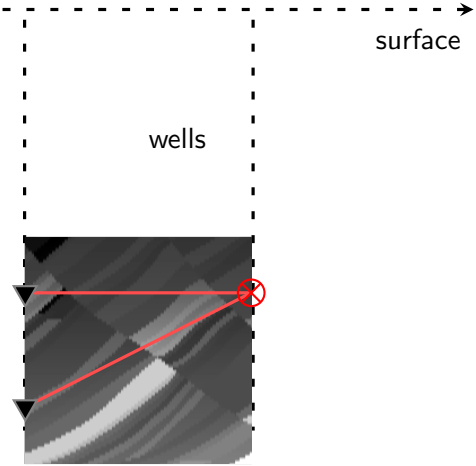
Seismic While Drilling (SWD)

- Two simultaneous sources:
  - An uncontrollable drill-bit source always emitting random signals.
  - Additional controllable active source used to improve illumination.
- At receivers:
  - Unknown drill-bit source signature prevents use of conventional imaging methods.
  - Active signals are contaminated by drill-bit noise.



Earth Model

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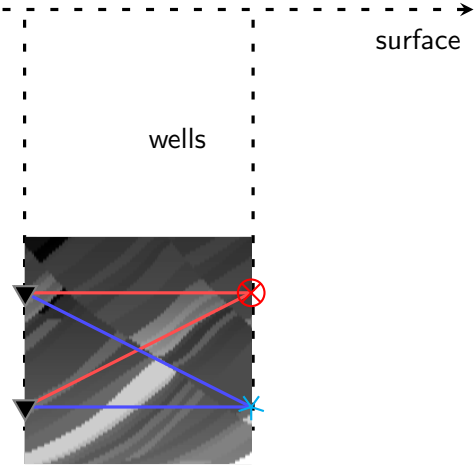

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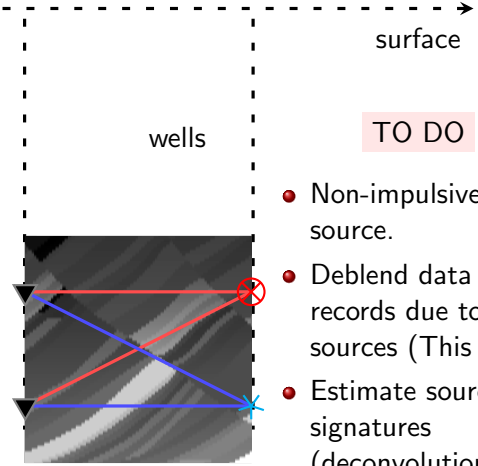


Earth Model

TO DO

- Non-impulsive active source.
- Deblend data into records due to individual sources (This Talk).
- Estimate source signatures (deconvolution).

Overview

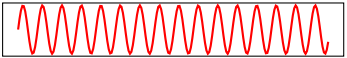
1. Independent Component Analysis (ICA)
2. Simple Instantaneous Mixtures
3. Seismic Convolutive Mixtures
4. Conclusions

Next Section

1. Independent Component Analysis (ICA)
2. Simple Instantaneous Mixtures
3. Seismic Convolutive Mixtures
4. Conclusions

Mixture Model: **H** Mixes B and S

$$\begin{pmatrix} B(\omega) \\ S(\omega) \end{pmatrix} \begin{matrix} \rightarrow \text{Noise Source} \\ \rightarrow \text{Active Source} \end{matrix}$$



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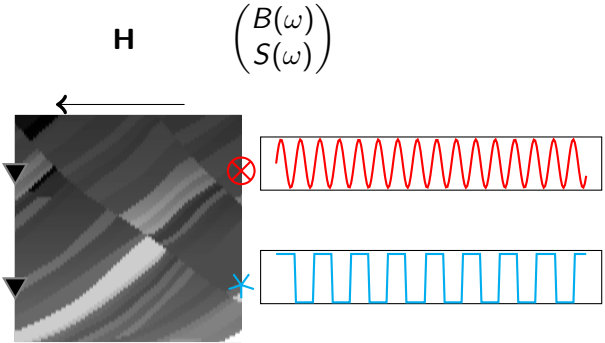
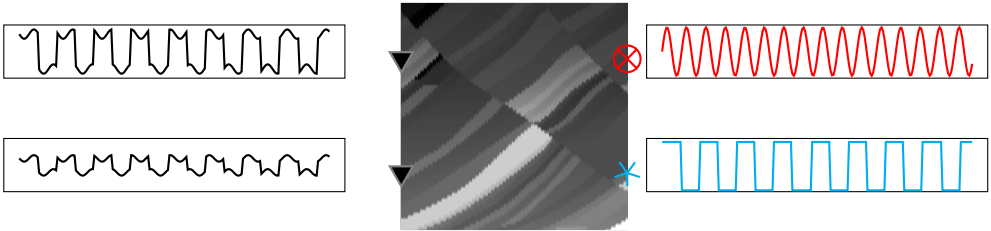


Mixture Model: **H** Mixes B and S

Mixing

$$\text{Data Vector} \quad \begin{pmatrix} D_1(\omega) \\ D_2(\omega) \end{pmatrix} = \mathbf{H} \begin{pmatrix} B(\omega) \\ S(\omega) \end{pmatrix}$$



Earth Model

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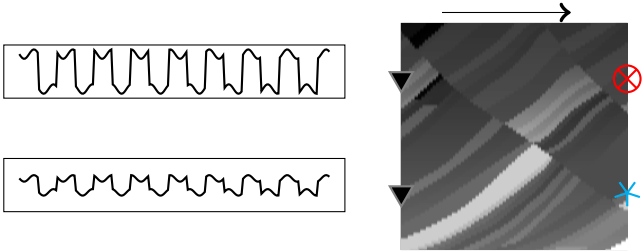
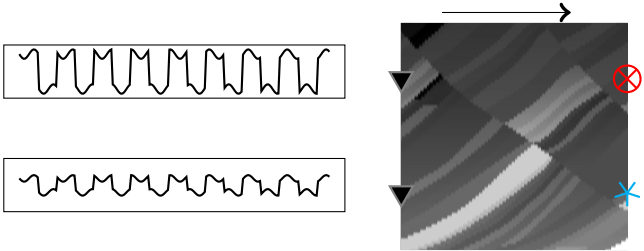


Earth Model

Mixture Model: Can We Find a **W**?

$$\mathbf{W}\begin{pmatrix} D_1(\omega) \\ D_2(\omega) \end{pmatrix} = \mathbf{W}\ \mathbf{H}\begin{pmatrix} B(\omega) \\ S(\omega) \end{pmatrix}$$

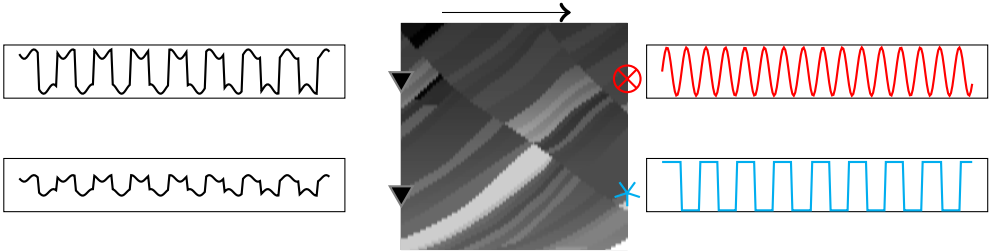


Earth Model

Mixture Model: Can We Find a **W**?

Unmixing

$$\mathbf{W}\begin{pmatrix} D_1(\omega) \\ D_2(\omega) \end{pmatrix} = \overset{\mathbf{I}}{\cancel{\mathbf{W}\ \mathbf{H}}} \begin{pmatrix} B(\omega) \\ S(\omega) \end{pmatrix}$$



Earth Model

Mixture Model

$$\mathbf{W}\begin{pmatrix} D_1(\omega) \\ D_2(\omega) \end{pmatrix} = \overset{\mathbf{I}}{\cancel{\mathbf{W}\ \mathbf{H}}} \begin{pmatrix} B(\omega) \\ S(\omega) \end{pmatrix}$$

Assumptions

- B and S must be statistically independent.
- B or S should be non-Gaussian.

Limitations

- B and S are recovered only up to a scaling factor.

Next Section

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During Mixing...

$$\begin{pmatrix} D_1(\omega) \\ D_2(\omega) \end{pmatrix} = \mathbf{H} \begin{pmatrix} B(\omega) \\ S(\omega) \end{pmatrix}$$

$$\longleftarrow$$

- `Gaussianity` increases (central limit theorem)
  - abs(`Kurtosis`) decreases
- `Dependency` and `Correlatedness` increases

During Mixing...

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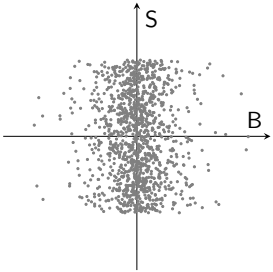
During Mixing. . .

Independent Source Signals



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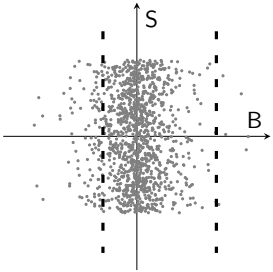
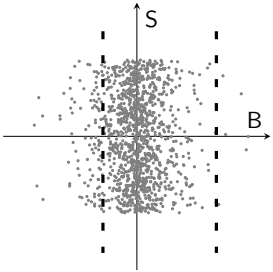


During Mixing...

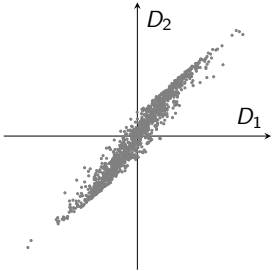
$$P(S|B) = P(S)$$
$$P(B,S) = P(B)P(S)$$

Independent Source Signals



During Mixing...

Observed Data



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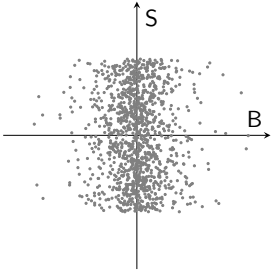

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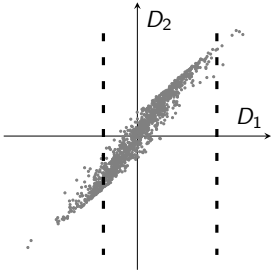

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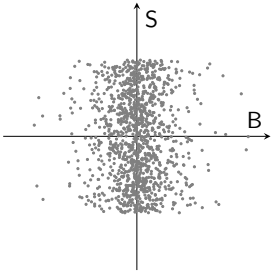

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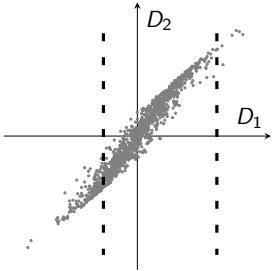
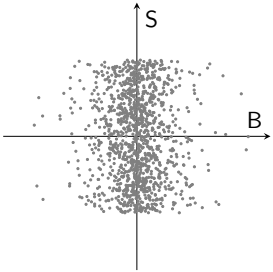


$$P(D_1, D_2) \neq P(D_1)P(D_2)$$
$$P(D_2|D_1) \neq P(D_2)$$

Independent Source Signals



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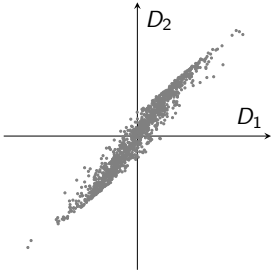

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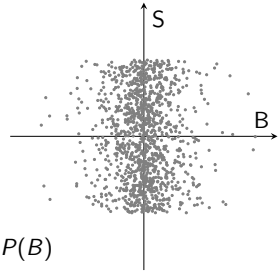
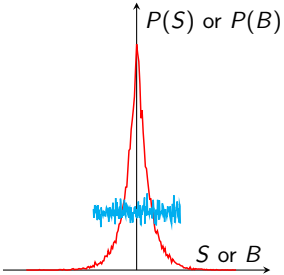


Marginal Densities



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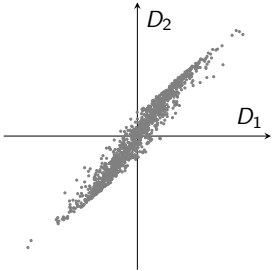

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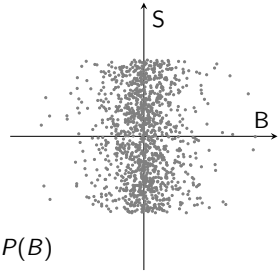

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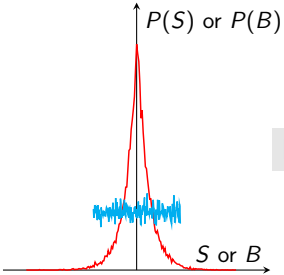

Kurtosis= -1.2

Kurtosis=3

During Mixing...

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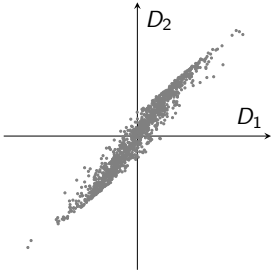

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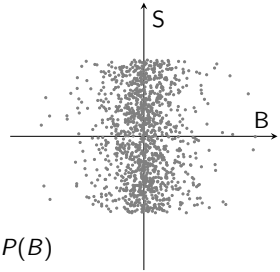

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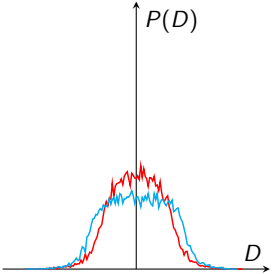

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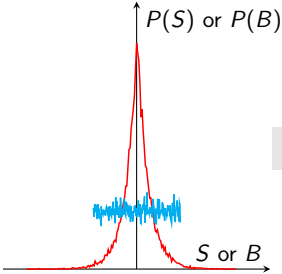

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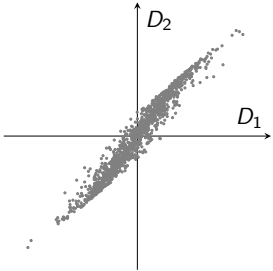

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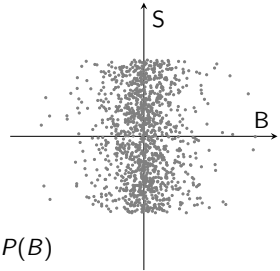

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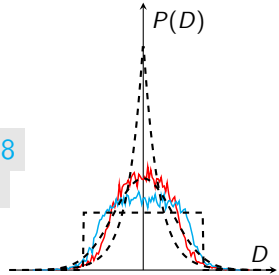
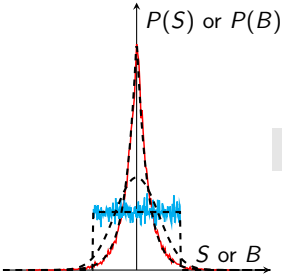


Kurtosis= -0.8

Kurtosis=0.5

Marginal Densities



Kurtosis= -1.2

Kurtosis=3

During Unmixing...

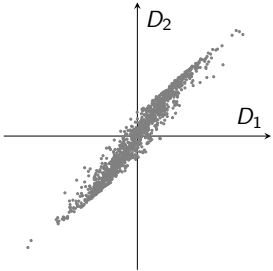
$$\mathbf{W}\begin{pmatrix} D_1(\omega) \\ D_2(\omega) \end{pmatrix} = \mathbf{W}\ \mathbf{H}\begin{pmatrix} B(\omega) \\ S(\omega) \end{pmatrix}$$

$$\longrightarrow$$

- Decrease `Gaussianity`
  - Increase abs(`Kurtosis`)
- Remove `Dependency` (ICA algorithm using Mutual Information)

During Unmixing...

Observed Data



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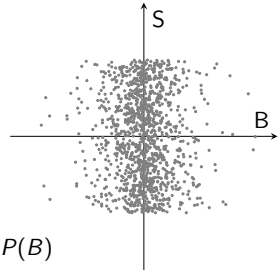

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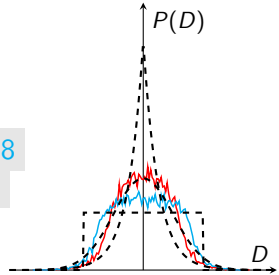

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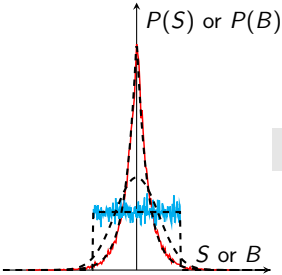

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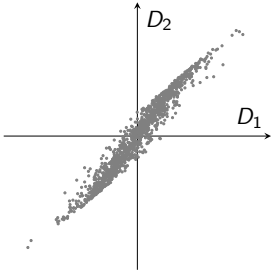
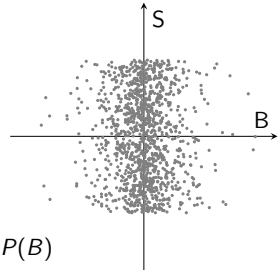
During Unmixing...

Observed Data



$$\mathrm{E}(D_1\, D_2) \neq \mathrm{E}(D_1)\mathrm{E}(D_2) = 0$$

Independent Source Signals



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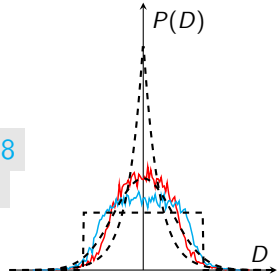

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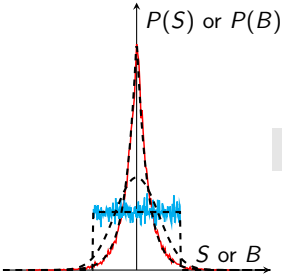

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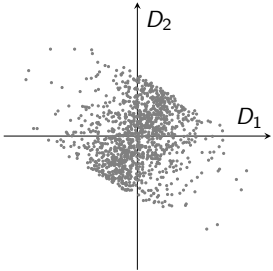
During Unmixing...

Whitened Data



$$\mathrm{E}(D_1\, D_2) = \mathrm{E}(D_1)\mathrm{E}(D_2) = 0$$

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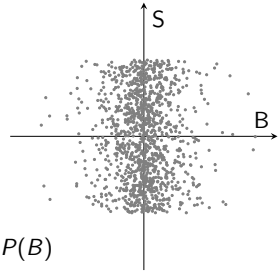

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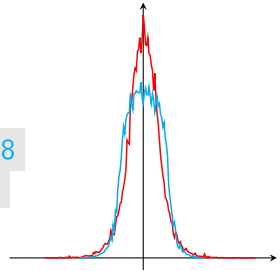
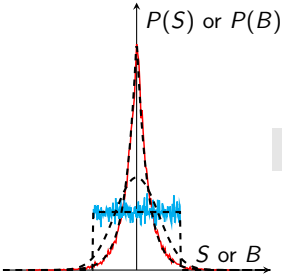


Kurtosis= -0.8

Kurtosis=0.5

Marginal Densities

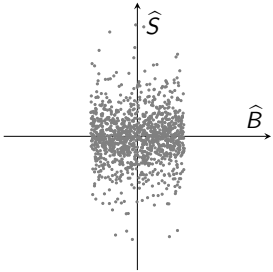


Kurtosis= -1.2

Kurtosis=3

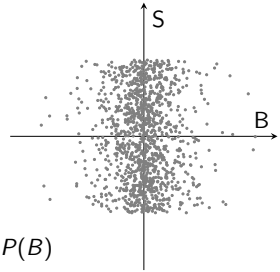
During Unmixing...

ICA Outputs

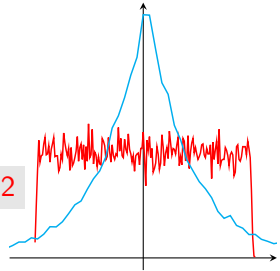


$$P(\widehat{B}, \widehat{S}) = P(\widehat{B})P(\widehat{S})$$

Independent Source Signals



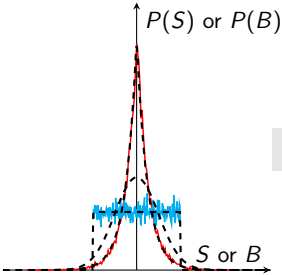
Marginal Densities



Kurtosis=3

Kurtosis= -1.2

Marginal Densities



Kurtosis= -1.2

Kurtosis=3

Mixture Model

$$\mathbf{W} \begin{pmatrix} D_1(\omega) \\ D_2(\omega) \end{pmatrix} = \mathbf{W} \ \mathbf{H} \begin{pmatrix} B(\omega) \\ S(\omega) \end{pmatrix}$$

Assumptions

- B and S must be statistically independent.
- B or S should be non-Gaussian.

Limitations

- B and S are recovered only up to a scaling factor.

Mixture Model

$$\mathbf{W}\begin{pmatrix} D_1(\omega) \\ D_2(\omega) \end{pmatrix} = \mathbf{W}\ \mathbf{H}\begin{pmatrix} B(\omega) \\ S(\omega) \end{pmatrix}$$

Assumptions

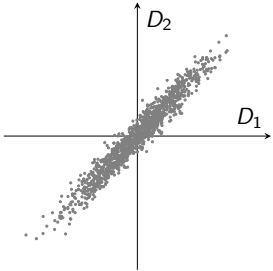
- B and S must be statistically independent.
- B or S should be non-Gaussian.

Limitations

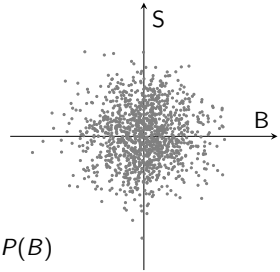
- B and S are recovered only up to a scaling factor.

Why Not a Gaussian Mixture

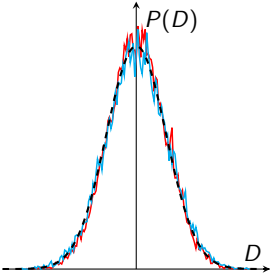
Observed Data



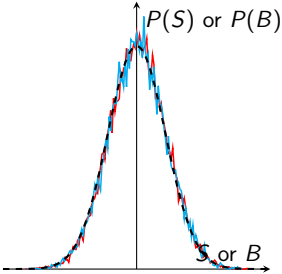
Gaussian Source Signals



Marginal Densities



Marginal Densities



Kurtosis=0
Kurtosis=0

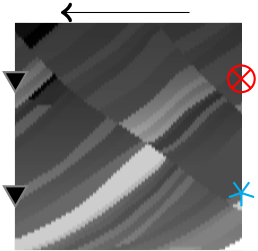
Kurtosis=0
Kurtosis=0

Next Section

1. Independent Component Analysis (ICA)
2. Simple Instantaneous Mixtures
3. Seismic Convolutive Mixtures
4. Conclusions

Seismic Convolutive Mixture Model

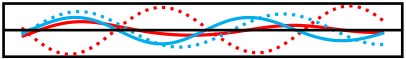
$$\begin{pmatrix} D_1(\omega) \\ D_2(\omega) \end{pmatrix} = \mathbf{H} \begin{pmatrix} B(\omega) \\ S(\omega) \end{pmatrix}$$



Earth Model

Seismic Convolutive Mixture Model

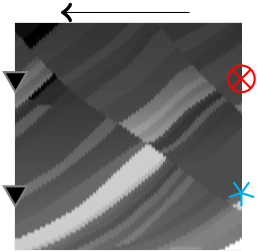
Green's Functions



$$\begin{pmatrix} D_1(\omega) \\ D_2(\omega) \end{pmatrix} = \mathbf{A}(\omega) \begin{pmatrix} B(\omega) \\ S(\omega) \end{pmatrix}$$

A's Resolution: $\frac{2\pi}{\tau}$

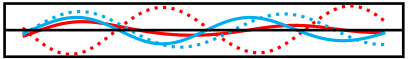
τ is Maximum Wave Propagation Time



Earth Model

Seismic Convolutive Mixture Model

Green's Functions



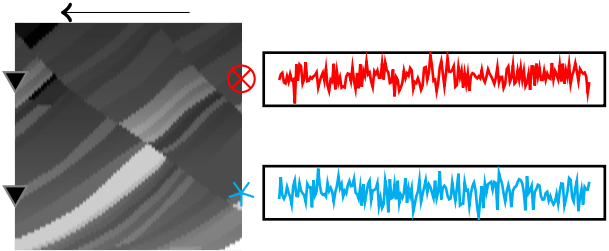
$$\begin{pmatrix} D_1(\omega) \\ D_2(\omega) \end{pmatrix} = \mathbf{A}(\omega) \begin{pmatrix} B(\omega) \\ S(\omega) \end{pmatrix}$$

A's Resolution: $\frac{2\pi}{\tau}$

τ is Maximum Wave Propagation Time

S and B Resolution: $\frac{2\pi}{T}$

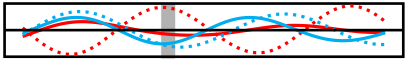
T is Total Record Time



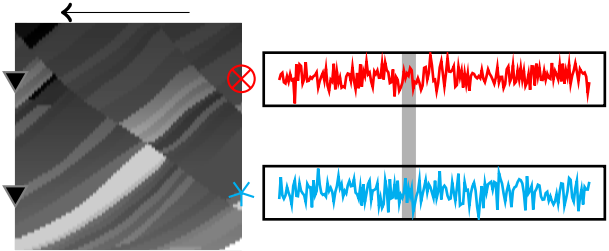
Earth Model

Seismic Convolutive Mixture Model

Green's Functions



$$\begin{pmatrix} D_1(\omega) \\ D_2(\omega) \end{pmatrix} = \mathbf{A}(\omega) \begin{pmatrix} B(\omega) \\ S(\omega) \end{pmatrix}$$

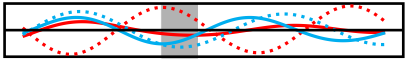


Scale Separation: $T >> \tau$

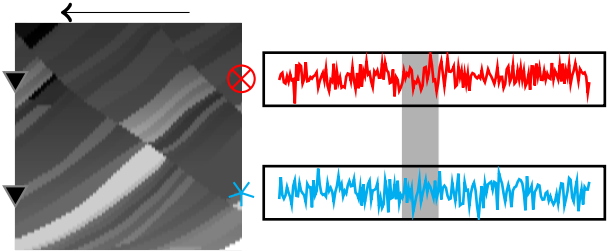
Earth Model

Seismic Convolutive Mixture Model

Green's Functions



$$\begin{pmatrix} D_1(\omega) \\ D_2(\omega) \end{pmatrix} = \mathbf{A}(\omega) \begin{pmatrix} B(\omega) \\ S(\omega) \end{pmatrix}$$

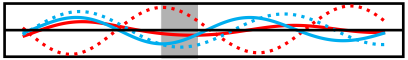


Scale Separation: $T >> \tau$

Earth Model

Seismic Convolutive Mixture Model

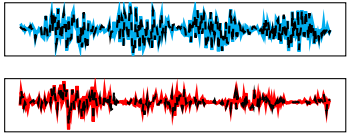
Green's Functions



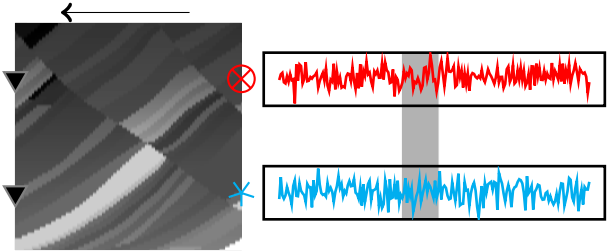
$$\begin{pmatrix} D_1(\omega) \\ D_2(\omega) \end{pmatrix} = \mathbf{A}(\omega) \begin{pmatrix} B(\omega) \\ S(\omega) \end{pmatrix}$$

$A_1 S$

$A_1 B$



$T \approx 10^4 \tau$

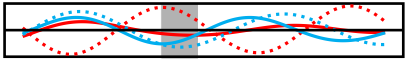


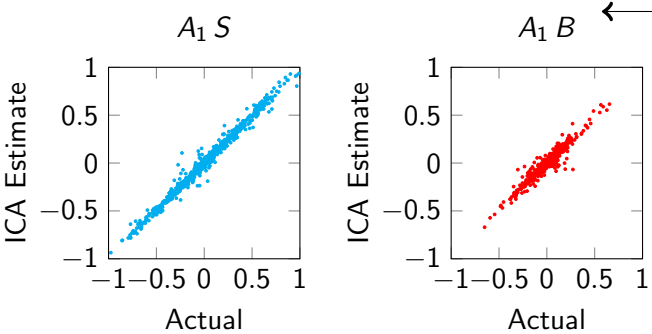
Earth Model

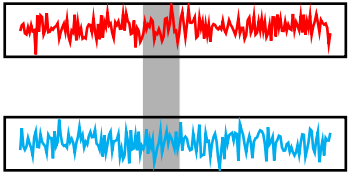
Seismic Convolutive Mixture Model

Optimal Freq. Bin Width
0.04 Hz (512 Samples)

Green's Functions



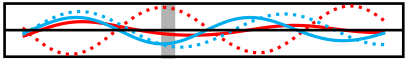


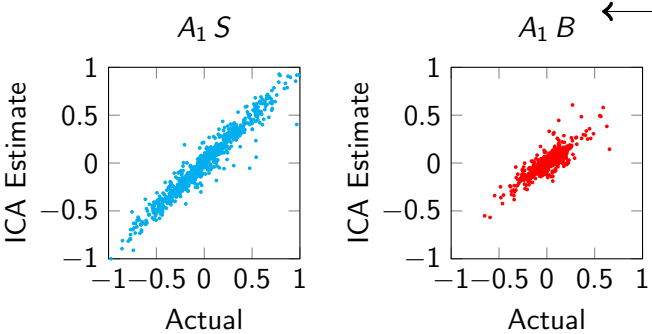


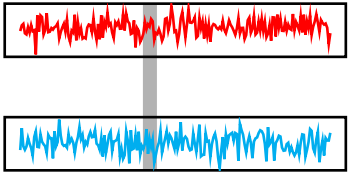
Seismic Convolutive Mixture Model

Freq. Bin Too Narrow
0.01 Hz (128 Samples)

Green's Functions



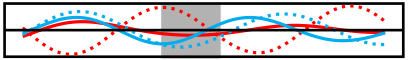


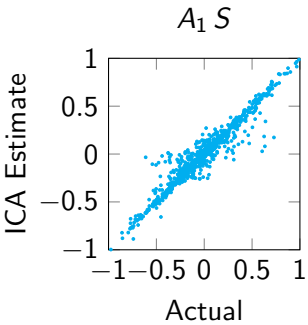


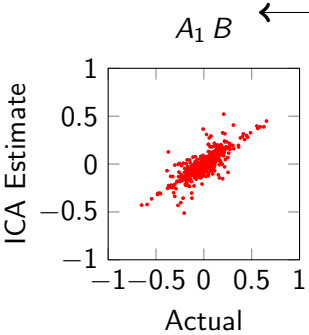
Seismic Convolutive Mixture Model

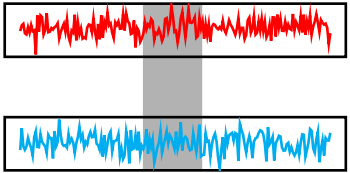
Freq. Bin Too Wide
0.16 Hz (2048 Samples)

Green's Functions



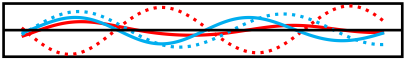


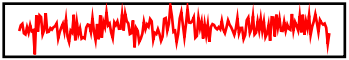


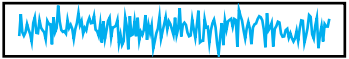


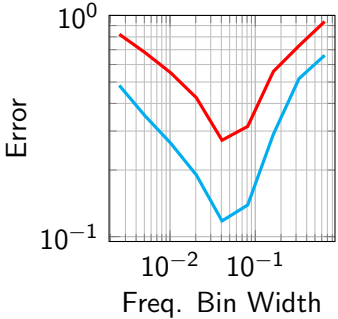
Seismic Convolutive Mixture Model

Green's Functions









Next Section

1. Independent Component Analysis (ICA)
2. Simple Instantaneous Mixtures
3. Seismic Convolutive Mixtures
4. Conclusions

ICA for Instantaneous Mixtures

- ICA is a simple model that estimates independent components from their mixture.
- Either of the two components must be non-Gaussian.

ICA for Seismic Convolutive Mixtures

- Deblending of seismic records due to simultaneous random sources is possible when the record time T is much larger than propagation time τ.
- As an example, during SWD experiments we need T to be atleast $10^4\,\tau$.
- ICA applicability to seismic shot records is fully and data driven and assumes a controllable non-Gaussian active source.

Acknowledgements

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- The authors thank Ioan Alexandru Merciu and Remus Gabriel Hanea from Statoil for their comments.
- LD is also funded by AFOSR grants FA9550-12-1-0328 and FA9550-15-1-0078, ONR grant N00014-16-1-2122, NSF grant DMS-1255203.

Thank You! Folks! Any Questions?

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